SMART IOT BASED IRRIGATION SYSTEM

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Abstract

This project presents a smart irrigation system that automatically controls watering based on real-time soil moisture, temperature, humidity, rain condition, and water level. The system is powered by solar energy, ensuring continuous operation even in remote and power-cut areas. LCD displays sensor data, OLED shows weather/API information and pump decisions, while a web/app interface enables remote monitoring and control. The system improves water efficiency, reduces manual effort, and supports smart farming.

Problem Statement:

- Manual irrigation causes water wastage and inefficiency.
- Lack of real-time monitoring and decision-making support.
- Power failure in rural and remote locations affects irrigation.
- Over-irrigation damages soil health and crop quality.
- No centralized platform for monitoring field conditions.
- Farmers need a reliable, low-cost automated solution.

Methodology

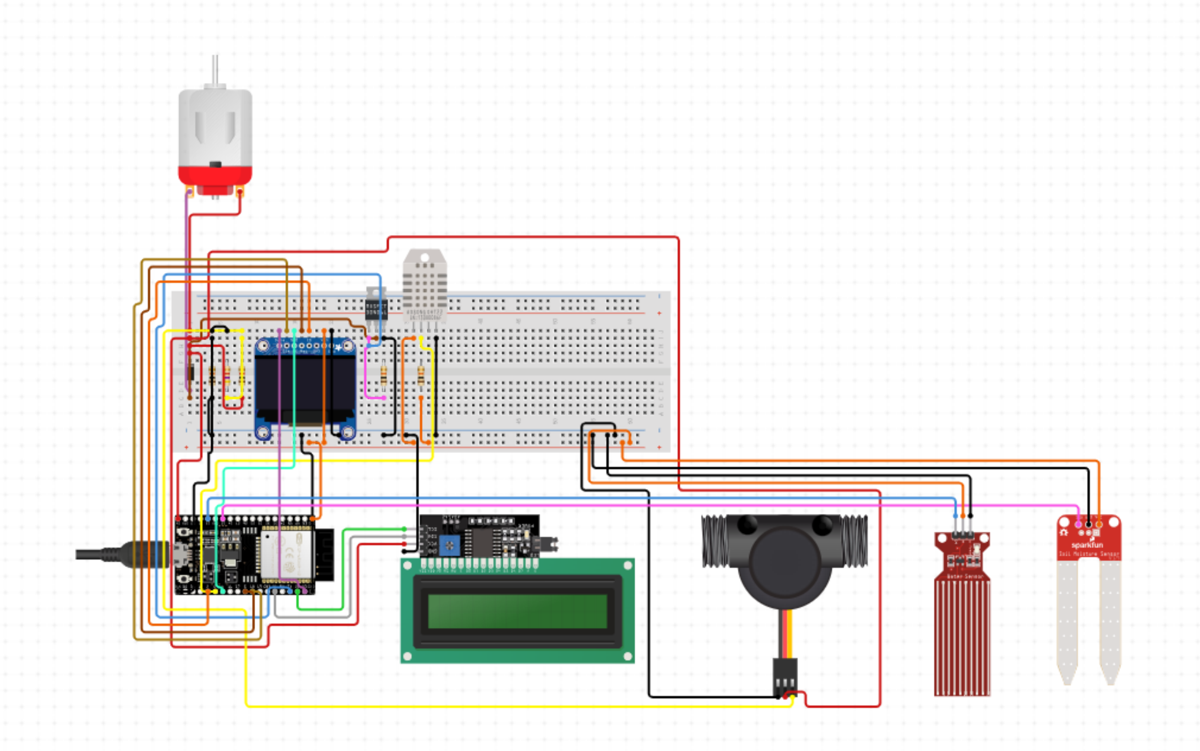
Sensors collect field data → ESP32 processes inputs → Weather API checked → Decision taken → Pump controlled via relay → Data displayed → Remote control available → Solar ensures power reliability.

SMART IOT BASED IRRIGATION SYSTEM

Features

- Automatic irrigation based on soil moisture.
- Rain and weather-aware irrigation decision.
- Water level based safety to avoid pump dry run.
- Dual Display:
    - LCD → Sensor values.
    - OLED → API data & pump decisions.
- Remote monitoring & control through app/web.
- Continuous real-time monitoring.
- Solar powered uninterrupted working.
- Alerts and fault protection.
- Simulation + Hardware implementation.

Circuit Diagram



SMART IOT BASED IRRIGATION SYSTEM

HARDWARE IMPLEMENTATION

Components Used

- ESP32 Microcontroller
- Soil Moisture Sensor
- DHT11/DHT22 (Temperature & Humidity)
- Rain Sensor / API Integration
- Water Level Sensor
- Relay Module
- Water Pump
- OLED Display
- LCD 16×2 Display
- Solar Panel + Battery Setup
- Connecting wires & power components

SOFTWARE IMPLEMENTATION

- Arduino IDE Programming
- Embedded C / C++ Code Development
- Firebase / Blynk Platform for Remote Monitoring
- Weather API Integration
- Data processing & decision logic implementation
- OLED + LCD display programming
- Simulation using Wokwi / Fritzing

9. TECHNOLOGY STACK

SMART IOT BASED IRRIGATION SYSTEM

Hardware: ESP32, Sensors, Relay, Pump, Solar Panel
Programming: C / C++ (Arduino IDE)
IoT Platform: Wi-Fi, Firebase / Blynk
Dashboard: Web/App monitoring
Simulation: Wokwi / Fritzing

Component	Module Pin	ESP32 Pin	Power Supply
LCD 16×2 (I2C)	**SDA**	**GPIO 21**	**5V**
LCD 16×2 (I2C)	**SCL**	**GPIO 22**	**5V**
OLED Display	**SDA**	**GPIO 21**	**3.3V or 5V**
OLED Display	**SCL**	**GPIO 22**	**3.3V or 5V**
Soil Moisture Sensor	**AO (Analog)**	**GPIO 34**	**3.3V or 5V**
DHT11/DHT22	**Data**	**GPIO 4**	**3.3V**
Rain Sensor	**Digital OUT**	**GPIO 32**	**5V**
Water Level Sensor	**Digital OUT**	**GPIO 33**	**5V**
Relay Module	**IN**	**GPIO 26**	**-**
Relay Module	**VCC**	**-**	**5V**
Relay Module	**GND**	**-**	**GND**

SMART IOT BASED IRRIGATION SYSTEM

Component	Module Pin	ESP32 Pin	Power Supply
Water Pump	-	Relay Output	External 5-12V
ESP32 Power	VIN/5V	-	5V USB or Buck
General Sensors	VCC/GND	-	3.3V/5V per spec

OBJECTIVES

- Automate irrigation using real-time environmental data.
- Reduce water usage through smart decision logic.
- Ensure reliable operation using solar power.
- Provide remote monitoring and control facility.
- Display field data and decision transparency.
- Improve farming efficiency and productivity.

Business Potential

- High demand in agriculture & smart farming
- Useful for farmers, greenhouses, NGOs & Govt schemes
- Low-cost, scalable, rural-friendly solution
- Can support subscription-based services
- Works effectively in power-scarce areas

AI / Data Intelligence Scope

SMART IOT BASED IRRIGATION SYSTEM

- AI-based irrigation optimization
- Crop yield prediction & fault detection
- Water usage analytics & trends
- Smart farming dashboard & insights

CONCLUSION (Updated Brief)

The Smart IoT Based Solar Powered Automatic Irrigation System provides efficient, automatic, and reliable irrigation while reducing water wastage and manual effort. It is practical, affordable, and highly suitable for rural and remote farming areas. With future AI integration and advanced analytics, even more automation can be achieved, making this a complete smart farming ecosystem. Overall, the proposed system delivers a low-cost, scalable, and full end-to-end smart agriculture solution.